

Dr. Babasaheb Ambedkar Technological University
Department of Electronics and Telecommunication Engineering

Supplementary Semester Examination 2019

Subject: Elective-I Electronic Measurements and Instruments (BET405B)

Semester : IV

Date: 08/06/2019

Time:

Maximum marks : 70

Instruction to Students

1. Assume suitable data wherever necessary & state it clearly.
2. Q.1 is compulsory and carry 10 Marks whereas Q.2 to Q.7 carries 12 Marks each.
3. Attempt any five question from Q.2 to Q.7.
4. Necessary data is given in respective question. If such data is not given, it means that the knowledge of that part is Examination.
5. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

QUE 1)

[10]

- 1) RF heterodyne wave analyzer operated in range of
i) 20 Hz-20KHz ii) 10KHz-18MHz iii) 500KHz-1GHz
iv) none of these
- 2) The expected value of voltage to be measured is 150V .however
measurement gives a value of 149 V. The absolute error is _____
i) +1 ii) -1 iii) 0 iv) +2
- 3) In digital frequency meter input signal control enabling and disabling
of main gate in _____ measurement
i) Period ii) time interval iii) frequency iv) all of these
- 4) Focusing anode is provided with positive voltage about
i) 2-10V ii) 2kV-10kV iii) 40-500V iv) none of these
- 5) A 30 Cm scale has 30 uniform divisions. $\frac{1}{20}^{\text{th}}$ of a scale division can
be estimated . The resolution of the scale is _____ mm.

- QUE 2)** A) Explain the term : accuracy, precision, Dynamic error [6]
 B) Explain RF voltage measurement using vector voltmeter [6]
- QUE 3)** A) Explain in details various sources of error. [6]
 B) Explain working principle of Q- Meter [6]
- QUE 4)** A) Explain principle of operation of digital frequency meter. [4]
 B) Define : arithmetic mean, average deviation , standard deviation [3]
 C) What is meant by standards ? What are the different types of standards of measurement [5]
- QUE 5)** A) Explain in details with working of Digital storage Oscilloscope [6]
 B) Explain Direct comparison and Indirect comparison calibration method [6]
- QUE 6)** A) Explain digital FFT analyzer [6]
 B) Explain most commonly used receiver technique of network analyzer measurement system [6]
- QUE 7)** A) Explain typical automatic test system with necessary block diagram [6]
 B) Explain IEEE 488 standards.. [6]

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